NutriSafe AI

A Multi-Modal Explainable AI System for Personalized Pediatric Food Additive Risk Assessment

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WHY THIS PROJECT?

Food labels are difficult for many parents to interpret. Packaged foods may contain artificial colourants, flavour enhancers, preservatives and other additives listed using chemical names or E-Numbers.

Existing food-scanning apps mainly provide generic product information, rather than assessing food according to an individual child's health profile.

THE NEED

A practical way to translate a food label into an evidence-supported, personalised screening result for a child.

THE PROBLEM

01 COMPLEX LABELS
Chemical names and E-Numbers are difficult for non-specialists to interpret.

02 ADDITIVE INTERACTIONS
Checking ingredients individually may not capture potential interactions between multiple additives.

03 NO PERSONALISATION
Generic food assessments do not sufficiently consider a child's allergies, sensitivities, health conditions and dietary history.

04 LIMITED EXPOSURE VIEW
Recorded food information is not transformed into useful longitudinal exposure trends.

RESEARCH GAP

Existing solutions provide useful capabilities separately:

Food Databases
↓
OCR / Document Understanding
↓
Chemical & Graph Modelling
↓
Explainable AI

BUT...
There is a gap in integrating these capabilities into one personalised pediatric food-additive screening workflow.

NutriSafe AI integrates: ingredient extraction, chemical resolution, interaction modelling, child-specific risk scoring and explainable AI.

EXISTING SOLUTIONS vs NUTRISAFE AI

Feature	Existing Solutions	NutriSafe AI
Product / Barcode Lookup	✓	✓
Ingredient Scanning	Limited / Data dependent	✓
Child Health Profile	✗	✓
Additive Interaction Assessment	✗	✓
Estimated Cumulative Exposure	✗	✓
Explainable Decision	Partial / Limited	✓ SHAP
Pediatrician / Regulator Prototype	✗	✓

NutriSafe AI combines existing food information with AI-based interaction modelling and child-specific factors rather than replacing existing food databases.

HOW NUTRISAFE AI WORKS

01 CAPTURE
Food Label or QR / Barcode

02 EXTRACT
Image Processing + OCR / VLM
Extract ingredient information from the food label.

03 IDENTIFY
Chemical Entity Resolution
Ingredient names & E-Numbers → Standardised additive identifiers

04 ANALYSE
Chemical Knowledge Graph
Rule-Based Baseline vs R-GCN Interaction Model

05 PERSONALISE
Child Health Profile
Age • Allergies
Sensitivities
Health Conditions
Exposure

06 PREDICT
XGBoost Risk Model
0 – 100 Personalised Risk Score

07 EXPLAIN
SHAP Explainability
Shows the factors contributing to the result.

PERSONALISED RISK RESULT

LOW / MODERATE / HIGH CONCERN

58 /100

CONTRIBUTING FACTORS
- Detected additives
- Interaction evidence
- Allergen / health-condition relevance
- Estimated dietary exposure

EXPLAINABLE RESULT
"Why did NutriSafe AI produce this result?"
SHAP-based explanations identify the features contributing to the risk score.

TECHNOLOGY STACK

MOBILE APPLICATION	WEB DASHBOARDS	BACKEND	COMPUTER VISION / OCR
React Native / Flutter	React.js Tailwind CSS	Python FastAPI	OpenCV OCR / VLM

AI / MACHINE LEARNING	KNOWLEDGE & DATA	CHEMICAL / NLP	OPTIONAL RECOMMENDATION
R-GCN PyTorch Geometric XGBoost SHAP	Neo4J PostgreSQL	Chemical Dictionary Biomedical NLP	FAISS (Lower Priority)

RESEARCH CONTRIBUTION

What we evaluate

Rule-Based Interaction Model **VS** R-GCN Interaction Model

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Personalised Risk Scoring + SHAP-Based Explainability

Research Focus
Can graph-based interaction modelling provide better additive-interaction screening than a rule-based baseline, while combining interaction evidence with child-specific factors to produce an explainable personalised result?

STAKEHOLDER MODULES

PARENTS (Primary Mobile Application)
- Scan food labels
- Create child profile
- View personalised risk
- Understand contributing factors
- Track scan history

PEDIATRICIANS (Secondary Prototype)
- View logged products
- View exposure history
- Review factors behind flagged products

FOOD-SAFETY AUTHORITIES (Secondary Prototype)
- View frequently flagged additives
- Explore aggregated market-level trends
- Use de-identified demonstration data

UI DESIGN (Consumer Mobile Application)

01 HOME / SCAN
NutriSafe AI
Check food before you choose.
Scan Food Label
Scan Barcode

02 CHILD PROFILE
Emma, 7 years
Allergies: Milk, Food colouring
Sensitivities: Preservatives
Health Conditions: None
Save Profile

03 SCAN RESULT
MODERATE CONCERN
58 /100
Contributing Factors: Detected additives, Interaction evidence, Child profile relevance, Exposure history
View Explanation

04 EXPLANATION
Why this result?
Risk Factors: Additive Interaction, Health Relevance, Previous Exposure
Explanation: The result is influenced by detected additives, interaction evidence and child-specific factors.
View Scan History

05 EXPOSURE HISTORY
Exposure History
Estimated Dietary Exposure

Product	Date	Concern
Product A	Mon	Low
Product B	Tue	Moderate
Product C	Thu	Moderate
Product D	Sat	Low

Trend (Last 7 Days)

EXPECTED OUTCOME

- Extract ingredient information from food labels
- Identify standardised chemical additives
- Analyse potential additive interactions
- Personalise results according to child-specific factors
- Track estimated dietary exposure over time
- Explain why a product received its risk category

IMPORTANT DISCLAIMER

NutriSafe AI is designed for food-safety screening and decision support.
It does not diagnose medical conditions, measure biological chemical concentrations, or replace professional pediatric medical advice.

Thank you!